Q1:

What is Linux? Explain any five advantages of using Linux.

Linux is a Unix-based open-source operating system that manages hardware, software, and resources just like other operating systems, such as Windows and macOS.

Here are five advantages of using Linux:

1- It is open source, which means its source code is available for anyone to view, modify, and improve. Most Linux distributions are free to use, so individuals and companies can reduce software costs.
2- It has a powerful permission system, regular security updates, and fewer common malware attacks compared to some other operating systems. This is one reason Linux is known for strong security and popular for servers.
3- It is Lightweight and fast, so it runs smoothly on both modern and older hardware. Many Linux distributions use fewer system resources, thereby improving performance.
4- It provides powerful command-line tools, scripting support, package managers, and compatibility with tools like Docker, Kubernetes, Git, Jenkins, and cloud platforms. This makes it very useful for software development and DevOps work.
5- It has a large global community. Many users, developers, and companies contribute tutorials, forums, documentation, and solutions, so it is easy to find help when learning or troubleshooting.

Q2:

Explain the difference between Open Source Software and Proprietary Software with suitable examples.

Open Source Software:

Open source software is software whose source code is available to the public. Users can view, use, modify, and share it in accordance with its license. It is usually free or low-cost.

Examples of open source software include Linux, Mozilla Firefox, LibreOffice, and VLC Media Player.

Proprietary Software:

Proprietary software is software owned by a company or individual. Its source code is not publicly available. Users usually need to purchase a license and cannot modify or share the software freely.

Examples of proprietary software include Microsoft Windows, Microsoft Office, Adobe Photoshop, and macOS.

Q3:

Write the Linux commands to perform the following tasks

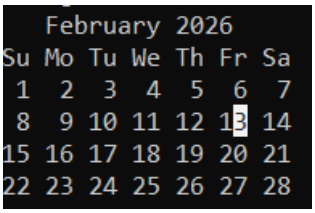
- Create a directory named training
  - *mkdir training*
- Create two files inside it
  - *cd training*
  - *touch file1 file2*
- Display all files
  - *ls*
- Rename one file
  - *mv file1 file3*
- Delete the other file

- *rm file2*

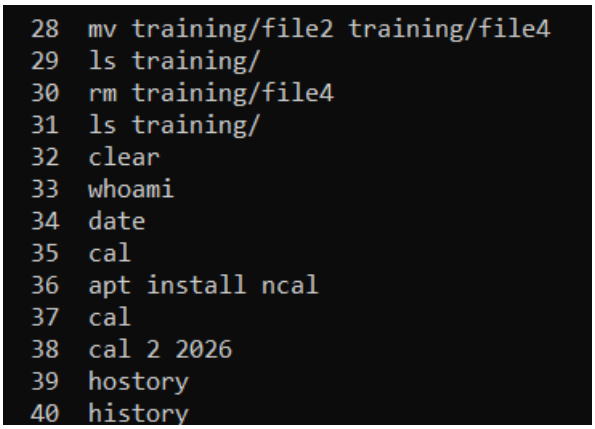
Q4:

Explain the purpose of the following Linux commands with one example each

- clear
  - This command is used to clean the terminal screen. It removes the previous commands and output from view.
- whoami
  - This command shows the username of the currently logged-in user.
  - **Example:** root
- date
  - This command displays the current date and time of the system.
  - **Exapmle:** Fri Jul 10 01:25:41 CEST 2026
- cal
  - This command displays the calendar of the current month. We can also show a specific month and year.
  - **Example:** cal 2 2026



- history
  - This command shows the list of commands that the user has previously entered in the terminal.
  - **Example:** history



Q5:

Write Linux commands to perform the following:

- Display the current date and time
  - *date*
- Display the current logged-in user
  - *whoami*
- Display the current working directory

- *pwd*
- Create a new folder named project
  - *mkdir project*
- Create an empty file named demo.txt
  - *touch demo.txt*

Q6:

Research any *three Linux commands* that were not covered during the training and explain their purpose with an example.

1. wc (Word Count) command
   - The wc command stands for word count. It counts the number of lines, words, and characters in a file. This command is useful for quickly checking the size or length of a text file.
   - Example: wc file3
   - Output: `1 14 66 file3` It shows the file contains 1 line, 14 words, and 66 characters

2. ln Command
   - This command is used to create links between files. It can create a hard link or a symbolic link.
   - **Soft Link:** A symbolic link works like a shortcut to another file or directory.
   - **For Examle:** ln -s file3.txt shortcut.txt

     ll

     ```
     66 Jul 10 10:01 file3
      9 Jul 10 10:10 shortcut.txt -> file3.txt
     ```

   - **Hard Link:** A hard link is another name for the same file on the disk. Both the original file and the hard link point to the same data. If the content in one file is changed, the same changes will appear in the other file.
   - **For Example:** ln file2.txt hardlink.txt

     ```
     46 Jul 10 10:17 file2.txt*
     66 Jul 10 10:01 file3*
     46 Jul 10 10:17 hardlink.txt*
     ```

   - When I made changes in hardlink.txt, the same changes were reflected in file2.txt
- **Note:** Even if the original file is deleted, the hard link will still work because the actual data remains on the disk as long as at least one hard link exists.

3. Stat command
   - This command is used to display detailed information about a file or directory. It shows information such as file size, permissions, owner, and last modified time.
   - For Example: stat hardlink.txt

     ```
       File: hardlink.txt
       Size: 68              Blocks: 0          IO Block: 4096   regular file
     Device: 0,71    Inode: 129197014310246836  Links: 2
     Access: (0777/-rwxrwxrwx)  Uid: ( 1000/    amna)   Gid: ( 1000/    amna)
     Access: 2026-07-10 10:18:52.695844400 +0200
     Modify: 2026-07-10 10:18:52.695844400 +0200
     Change: 2026-07-10 10:19:06.305168700 +0200
      Birth: -
     ```

Q7:

Explain the purpose of the following Linux directories:

- /home
  - This directory stores personal files and settings for users. Each user usually has their own folder inside /home.
  - **Example:** /home/amna
- /etc

    - This directory contains system configuration files. Many important settings for services, users, networks, and applications are stored here.
    - **Example:** /etc/passwd
- /var
    - This directory stores files that change frequently while the system is running. It commonly contains logs, cache, mail, and temporary service data.
    - **Example:** /var/log
- /usr
    - This directory contains user programs, libraries, documentation, and shared resources. Many installed applications and commands are stored here.
    - **Example:** /usr/bin
- /tmp
    - This directory stores temporary files created by the system or applications. These files are usually temporary and may be automatically deleted.
    - **Example:** An application may store a temporary file in /tmp while it is running.

Q8:

What are Linux file permissions? Explain *Read*, *Write*, and *Execute* permissions with suitable examples.

Linux file permissions control who can access a file or directory and what actions they can perform. These permissions help protect files from unauthorized reading, editing, or execution.

Linux mainly uses three types of permissions: Read, Write, and Execute.

Read Permission:

Read permission allows a user to view the contents of a file. For a directory, read permission allows the user to list the files inside that directory.

Example: cat file2.txt

If the user has read permission, they can view the contents of file2.txt.

Write Permission:

Write permission allows a user to modify or change a file. It also allows the user to delete or add content in the file. For a directory, write permission allows creating, deleting, and renaming files within it.

Example: echo "Hello Linux" >> file3.txt

If the user has write permission, this command adds text to file3.txt.

Execute Permission:

Execute permission allows a user to run a file as a program or script. For a directory, execute permission allows a user to enter it using the cd command.

Example: ./script.sh

If the user has execute permission, they can run the script file script.sh.

Example of File Permission

-rwxrw-r--

This means:

- The owner has read, write, and execute permissions. rwx
- The group has read and write permission. rw-
- Others have read-only permission. r--

Q9:

Suppose you are unable to edit a file because of permission issues. Which commands would you use to identify and resolve the problem?

First, I would check the file permissions.

ls -l file2.txt

Output: -r--r--r-- 1 amna users 120 Jul 10 file2.txt

This shows whether the owner, group, and others have read, write, or execute permissions. If there is no write, it means write permission is missing, and that's why the user can't edit the file.

Then I will check which current user is logged in.

Whoami This helps confirm whether I am the file's owner or another user.

I can also use the stat command to see detailed information about the file, including owner, group, and permissions. *stat file2.txt*

If I am the file owner, I can add write permission.

chmod u+w file2.txt

Here, u+w means adding write permission for the user/owner.

If the file belongs to another user and I have administrator permission, I can change the ownership.

sudo chown amna file2.txt

If the file is a system file, I may need to edit it using sudo.

sudo nano file2.txt

Q10:

What steps would you follow if you accidentally forgot the location of a file in your Linux system?

1- If I forgot the location of a file in Linux, I would first try to find it with the *find* command. The find command searches for files and directories in a given location.

   find /home -name "file2.txt"

   This command searches for a file named file2.txt inside the /home directory.

2- If I do not know where the file is stored, I can search the whole Linux system.
sudo find / -name "file2.txt"
Here, / means search from the root directory. sudo helps avoid permission errors while searching system directories.
3- If I remember only part of the file name, I can use * as a wildcard.

*find /home -name "*file2*"*

This command searches for files that contain the word file2 in their name.

4- I can also use the locate command to find a file quickly.
locate file2.txt
If the result is not updated, I can update the database first:
sudo updated
5- If I worked with the file recently, I can check command *history*. This may show the commands I used to create, move, or edit the file.

Q11:

What is Git? Explain why version control is important in software development.

Git is a version control system used to track changes in files and source code. It helps developers save different versions of their work and return to an earlier version if needed.

Importance of version control:

Version control is important because it records the history of changes in a project. It helps developers see what was changed, when it was changed, and who made the change.

It also helps teams work together safely. If a mistake happens, developers can restore a previous working version of the project.

Q12:

Write the Git commands to perform the following:

- Initialize a Git repository
  - *git init*
- Add a file
  - *git add filename.txt* *(To add a single file)*
  - *git add .* *(To add all files)*
- Commit the changes
  - *git commit -m "commit message/comment"*
- Check the repository status
  - *git status*
- View the commit history
  - *git log*

Q13:

Research any three Git commands that were not covered during the training and explain their purpose.

1- ***git diff***
This command shows the differences between the working files and the last committed version. It helps developers see what changes have been made before adding or committing them.

2- ***git stash***
This command temporarily saves changes without committing them. It is useful when a developer wants to switch to another branch without losing current work.

To bring the changes back, we can use:
git stash pop

3- ***git revert***
This command is used to safely undo a previous commit. It creates a new commit that reverses the changes from an earlier commit, rather than deleting the commit history.
Example: *git revert commit_id*

Q14:

What is GitHub? Explain the difference between Git and GitHub.

GitHub is an online platform for storing, sharing, and managing Git repositories. It helps developers collaborate on projects, review code, track issues, and work together from different locations.

Difference Between Git and GitHub:

Git is a version control tool installed on the computer. It tracks file changes and lets the user save multiple versions of the project.

GitHub is a cloud-based platform for hosting Git repositories and sharing them online. It makes collaboration easier by providing features such as pull requests, issues, and project management tools.

Q15:

Explain the purpose of the following GitHub features:

Repository: A repository is a project folder on GitHub. It stores the project files, code, documentation, and Git history.

Example: A repository named devops-internship can contain HTML, CSS, JavaScript files, and a README file.

Branch: A branch is a separate version of a repository. It allows developers to work on new features or fixes without changing the main project directly.

Example: A branch named login-feature can be used to develop a login page separately.

Clone: It means copying a GitHub repository from the internet to the local computer.

Example: *git clone https://github.com/user/project.git*

Commit: A commit is a saved change in a repository. It records what was changed and creates a point in the project history.

Example: *git commit -m "Added homepage"*

Pull Request: A request to merge changes from one branch into another. It is commonly used for reviewing code before adding it to the main project.

Example: After completing work on the login-feature, a developer creates a pull request to merge it into the main branch.

Q16:

Write a Bash shell script that accepts your Name, College, and Course as input and displays the entered information.

The solution is uploaded to GitHub.

Q17:

Write a Bash shell script to check whether a given number is Even or Odd and display an appropriate message.

Solution:

```
#!/bin/bash

read -rp "Enter a number: " number

if (($number%2 == 0));

then

    echo "You have entered an even number."

else

    echo "You have entered an odd number."

fi
```

Q18:

Write a Bash shell script that accepts a filename from the user and checks whether the file exists. If the file exists, display "File Found"; otherwise, display "File Not Found".

```
#!/bin/bash

read -rp "Enter a filename: " filename

if [[ -f $filename ]]

then

    echo "File found"

else
```

echo "File not found"

fi

Q19:

Write a Bash shell script that accepts a filename from the user and checks whether the file exists. If the file exists, display "File Found"; otherwise, display "File Not Found".

1. **Docker**
   Docker is a DevOps tool used to create, run, and manage applications inside containers. A container includes the application code, libraries, and dependencies, so the application can run the same way on different systems.

   Docker works well with Linux because containers commonly run in Linux environments. It also works with Git, as developers can store application code and Dockerfiles in a Git repository. When the code is updated, Docker can be used to build and run the latest version of the application.

   Docker is popular because it speeds up, simplifies, and standardizes application deployment. It helps developers avoid the problem of “it works on my machine but not on another system.” Docker is also widely used because Docker Hub provides a large container image registry for sharing and using ready-made images.

2. **Jenkins**
   Jenkins is an open-source automation server used for Continuous Integration and Continuous Delivery (CI/CD). It helps automate tasks such as building code, running tests, and deploying applications.

   Jenkins works well with Linux because it can run on Linux servers and automate Linux commands or shell scripts. It also works with Git because Jenkins can connect to Git repositories, check for new commits, pull the latest code, and run a pipeline automatically.

   Jenkins is popular because it saves time by automating repeated development tasks. It also supports many plugins, making it flexible across different DevOps workflows. Jenkins can be used as a simple CI server or as a complete continuous delivery system.

   Conclusion
   Docker and Jenkins are both important DevOps tools. Docker helps package and run applications in containers, while Jenkins helps automate building, testing, and deployment. Both tools work closely with Linux and Git in real-world DevOps projects.

Repository Link:

https://github.com/amna85shahid/linux-git-practice/

Commands Screenshots

Linux:

```
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode# mkdir training
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode# cd training/
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/training# touch file1 file2
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/training# mv file1 file3
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/training# ls
file2  file3
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/training# rm file2
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/training# ls
file3
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/training#
```

Scripts Execution:

```
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode# mkdir linux-git-practice
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode# cd linux-git-practice/
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/amna8/CareerByteCode/linux-git-practice/.git/
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# mkdir scripts linux-practice screenshots
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# touch README.md
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# touch linux-practice/linux_commands.md
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# vi student_info.sh
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# chmod 777 student_info.sh
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# ./student_info.sh
Enter your Name: Amna
Enter your College: DSV
Enter your Course: Computer Sciences

============================
      Entered Information
============================
Name    : Amna
College : DSV
Course  : Computer Sciences
============================
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# vi even_odd.sh
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# chmod u+w+x even_odd.sh
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# ./even_odd.sh
Enter a number: 7
You have entered an odd number.
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# vi file_exist.sh
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# chmod +x file_exist.sh
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# ./file_exist.sh
Enter a filename: README.md
File found
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# ./file_exist.sh
Enter a filename: file2
File not found
```

Git:

```
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# git add linux-practice/linux_commands.md
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# git commit -m "Add Linux command practice"
[main e294969] Add Linux command practice
 1 file changed, 151 insertions(+)
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
root@DESKTOP-BH2VINA:/mnt/c/Users/amna8/CareerByteCode/linux-git-practice# git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 986 bytes | 58.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/amna85shahid/linux-git-practice.git
   9765fd9..e294969  main -> main
branch 'main' set up to track 'origin/main'.
```